Corollary 3.5. *Every commutative n-ring of prime characteristic p is actually a p^k-ring for some $k \geq 1$. Namely, k is the* lcm *of all $d \geq 1$ with $p^d - 1 \mid n - 1$.*

Proof. Let A be a commutative n-ring of characteristic p. We have seen in Proposition 3.2 that A embeds into a product of fields of the form $\mathbb{F}_{p^d}$ with $p^d - 1 \mid n - 1$. But then $d \mid k$ and therefore $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^k}$ is also a p^k-ring. □

Remark 3.6. If n, p, k are as in Corollary 3.5, it is not true that, conversely, every p^k-ring of characteristic p is an n-ring, since we do not necessarily have $p^k - 1 \mid n - 1$. The smallest counterexample is $n = 22$ and $p = 2$. Then $k = \mathrm{lcm}(1, 2, 3) = 6$. Thus, $\mathbb{F}_{2^6}$ is an example of a 2^6-ring that is not a 22-ring.

We can turn any ring into an n-ring as follows.

Definition 3.7. If R is a ring, we define

$$R_{[n]} \coloneqq R/\langle x^n - x : x \in R\rangle.$$

This is clearly an n-ring with a surjective homomorphism $R \to R_{[n]}$. The construction is universal: If A is an n-ring and $R \to A$ is a homomorphism of rings, then it lifts uniquely to a homomorphism of n-rings $R_{[n]} \to A$. Therefore, we may call $R_{[n]}$ the universal n-ring quotient of R.

Example 3.8. Let us compute the universal n-ring quotient of $\mathbb{Z}$. We have

$$\mathbb{Z}_{[n]} \coloneqq \mathbb{Z}/\langle z^n - z : z \in \mathbb{Z}\rangle = \mathbb{Z}/\gcd(z^n - z : z \in \mathbb{Z})\mathbb{Z}.$$

Let c be the gcd in question. Of course, $c > 0$. Since $\mathbb{Z}_{[n]}$ is an n-ring and hence reduced, c is square-free. Let $p \mid c$ be a prime factor. This means $z^n \equiv z \bmod p$ for all $z \in \mathbb{Z}$. Equivalently, $\mathbb{F}_p$ is an n-ring, which means $p - 1 \mid n - 1$ by Remark 3.1. Therefore, c is the product of all primes p with $p - 1 \mid n - 1$. It follows that if A is an n-ring, then $c = 0$ holds in A, since the unique homomorphism $\mathbb{Z} \to A$ lifts to a homomorphism $\mathbb{Z}/c\mathbb{Z} = \mathbb{Z}_{[n]} \to A$. In other words, the characteristic of A divides c.

Corollary 3.9. *Every n-ring is a finite product of n-rings of prime characteristic.*

Proof. Let A be an n-ring. By Example 3.8 its characteristic is $p_1 \cdots p_s$ with distinct prime numbers $p_1, \ldots, p_s$. Then the Chinese Remainder Theorem (which holds for all rings and sequences of pairwise coprime two-sided ideals) tells us that the canonical homomorphism $A = A/p_1 \cdots p_s A \to A/p_1 A \times \cdots \times A/p_s A$ is an isomorphism. □

As a consequence, the commutativity problem can be reduced to prime characteristic. We need to convince ourselves, though, that there is also an equational proof of this reduction.

Remark 3.10. Even though it is tempting to say that every boolean ring has characteristic 2, this is not true: The trivial ring is a boolean ring of characteristic 1. Every non-trivial boolean ring has characteristic 2. But since the LEM is not available for equational proofs, we cannot (and do not need to!) decide if a given ring is trivial or not. There is no need to exclude the trivial ring. Therefore, in the following, we will not speak of rings of characteristic p, and instead only add $p = 0$ as an axiom (a shorthand for $p \cdot 1_A = 0$). Equivalently, we work with $\mathbb{F}_p$-algebras. Moreover, notice that in the proof of Corollary 3.9 the rings $A/p_i A$ do not necessary have characteristic p_i, since they could be trivial. They just satisfy $p_i = 0$. Of course, Corollary 3.9 remains true (in classical logic).

Lemma 3.11. *Let A be an n-ring. Let $p_1, \ldots, p_s$ be the prime numbers p with $p - 1 \mid n - 1$. Then there is an equational proof of $p_1 \cdots p_s = 0$ in A.*

Proof. We saw in Example 3.8 that $p_1 \cdots p_s$ is the gcd of all $z^n - z$ with $z \in \mathbb{Z}$. The argument used finite fields, which is not an equational concept, but if n is fixed we can ignore this and just compute the gcd of the numbers $z^n - z$ for $z = 1, 2, 3, \dots$ with the Euclidean algorithm until we reach $p_1 \cdots p_s$. It suffices to consider prime numbers for z. Then we use the extended Euclidean algorithm to write $p_1 \cdots p_s$ as a $\mathbb{Z}$-linear combination of the numbers $z^n - z$. These numbers vanish in A, so that $p_1 \cdots p_s$ vanishes as well. □

SageMath can do the computation for us for every value of n, see appendix A.2 for the code.

Example 3.12. Consider $n = 7$. The prime numbers p with $p - 1 \mid 7 - 1$ are $p = 2, 3, 7$. We find $\gcd(2^7 - 2, 3^7 - 3) = 42 = 2 \cdot 3 \cdot 7$ and $42 = -17(2^7 - 2) + (3^7 - 3)$. Hence, $2 \cdot 3 \cdot 7 = 0$ holds in any 7-ring. For $n = 3$, the primes are $2, 3$, and we already have $2^3 - 2 = 6 = 2 \cdot 3$.

Lemma 3.13. *Let $\mathcal{T}$ be an algebraic theory extending ring theory by some axioms, and let $p \in \mathbb{Z}$. If σ, τ are terms in $\mathcal{T}$ of the same arity such that $\mathcal{T}, p = 0 \vdash \sigma = \tau$, then there is a term u of the same arity with $\mathcal{T} \vdash \sigma = \tau + p \cdot u$.*

Proof. This is a simple induction on the structure of an equational proof as in Definition 2.1. We only show the most interesting one, that is rule (5). Assume that $\mathcal{T}$ and $p = 0$ prove the equation $\rho(\sigma_1, \dots, \sigma_n) = \rho(\tau_1, \dots, \tau_n)$ by means of $\sigma_i = \tau_i$ for all i. By induction hypothesis, there are terms $u_1, \dots, u_n$ such that $\mathcal{T}$ proves $\sigma_i = \tau_i + p \cdot u_i$. Then $\mathcal{T}$ proves the equation $\rho(\sigma_1, \dots, \sigma_n) = \rho(\tau_1 + p \cdot u_1, \dots, \tau_n + p \cdot u_n)$. Expanding the right hand side yields a term v with $\rho(\tau_1 + p \cdot u_1, \dots, \tau_n + p \cdot u_n) = \rho(\tau_1, \dots, \tau_n) + p \cdot v$, as desired. Formally, this requires an induction on the structure of ρ. □

Theorem 3.14. *Assume that for every $p \in \mathbb{P}$ with $p - 1 \mid n - 1$ there is an equational proof that every n-ring with $p = 0$ is commutative. Then there is an equational proof that every n-ring is commutative.*

Because of this result, we may restrict our attention to rings with $p = 0$ from now on.

Proof. Let A be an n-ring. Let $p_1, \dots, p_s$ be the set of $p \in \mathbb{P}$ satisfying $p - 1 \mid n - 1$. Then $p_1 \cdots p_s = 0$ holds in A by Lemma 3.11. If $x, y \in A$, by assumption $xy - yx = 0$ can be derived with the additional axiom $p_i = 0$. Hence, Lemma 3.13 yields $u_i \in A$ with $xy - yx = p_i \cdot u_i$. In classical logic, this step of the proof would be easier: just use that $[x]$ and $[y]$ commute in the quotient ring $A/p_i A$.

Therefore, it suffices to prove the following in equational logic: If $p_1, \dots, p_s$ are pairwise coprime integers and $x \in A$, $u_i \in A$ satisfy $x = p_i \cdot u_i$ for $i = 1, \dots, s$, then we can construct an element v with $x = p_1 \cdots p_s \cdot v$. The case $s = 1$ is easy. If $s = 2$, the extended Euclidean algorithm yields $q_1, q_2 \in \mathbb{Z}$ with $q_1 p_1 + q_2 p_2 = 1$. Then

$$x = q_1 p_1 x + q_2 p_2 x = q_1 p_1 p_2 u_2 + q_2 p_2 p_1 u_1 = p_1 p_2 (q_1 u_2 + q_2 u_1).$$

For the general case, we use induction and know $x = p_1 \cdots p_{s-1} \cdot v$ for some element v. Since $p_1 \cdots p_{s-1}$ and p_s are coprime, the case $s = 2$ yields an element w with $x = p_1 \cdots p_{s-1} p_s \cdot w$, and we are done. □

Example 3.15. Consider $n = 5$. The primes p with $p - 1 \mid 5 - 1$ are $2, 3, 5$, and $2^5 - 2 = 2 \cdot 3 \cdot 5$. Hence, $2 \cdot 3 \cdot 5 = 0$ holds in any 5-ring. Assume that there is a commutativity proof modulo each of these primes. Then we get elements u_1, u_2, u_3 with $xy - yx = 2u_1$, $xy - yx = 3u_2$, $xy - yx = 5u_3$. Hence, $xy - yx = 30(u_3 - u_1 + u_2) = 0$.

Remark 3.16. By Stone duality [St36], the category of boolean rings is anti-equivalent to the category of totally disconnected compact Hausdorff spaces (Stone spaces). There is a similar classification for n-rings, which is thus much more precise than Proposition 3.2. Namely, if $p^{d_1}, \dots, p^{d_s}$ are the powers of p with $p^{d_i} - 1 \mid n - 1$ and $k := \operatorname{lcm}(d_1, \dots, d_s)$, then the category of n-rings with $p = 0$ is anti-equivalent to the category of Stone spaces X with a continuous C_k-action such that $X = \bigcup_{i=1}^{s} X^{C_{k/d_i}}$. The proof of this anti-equivalence will appear elsewhere.

4. Reduction to prime powers

The goal of this section is to find an equational proof of Corollary 3.5.

Theorem 4.1. *Let $n > 1$ and let $p \in \mathbb{P}$ with $p - 1 \mid n - 1$. Let k be the* lcm *of all $d \geq 1$ such that $p^d - 1 \mid n - 1$. Then there is an equational proof that every n-ring with $p = 0$ is a p^k-ring. Moreover, if $p^k - 1 \mid n - 1$, the converse is also true.*

This is actually a meta-theorem. The proof is a method that enables us to write down, for every fixed pair (n, p), an equational proof that every n-ring with $p = 0$ is a p^k-ring. We will demonstrate this with several examples. Recall from Remark 3.10 why we write $p = 0$ instead of "characteristic p". The theorem is a generalization and better explanation of Morita's reduction results [Mo78] mentioned in the introduction. It also includes MacHale's observation in [Ma86] that $2^m + 1$-rings with $2 = 0$ are boolean as a special case.

Proof of Theorem 4.1. Since we merely want to show $x^{p^k} = x$, it is clear that only polynomials in one variable over $\mathbb{F}_p$ appear in the proofs, so that commutativity is not an issue here.

The first step is to use the Euclidean algorithm to compute the polynomial

$$g \coloneqq \gcd(f^n - f : f \in \mathbb{F}_p[T],\, \deg(f) < n) \in \mathbb{F}_p[T]. \tag{2}$$

This computation can be quite cumbersome, but see Remark 4.3 below for performance issues. Then one uses the extended Euclidean algorithm to write g as a linear combination of polynomials of the form $f^n - f$. We claim that

$$g \mid T^{p^k} - T. \tag{3}$$

The following proof is abstract and certainly not equational, but in practice, for every fixed pair (n, p), eq. (3) can just be verified via polynomial division. This why an equational proof still does exist.

The universal n-ring with $p = 0$ on one generator is the quotient $\mathbb{F}_p[T]_{[n]} = \mathbb{F}_p[T]/I$, where $I \coloneqq \langle f^n - f : f \in \mathbb{F}_p[T] \rangle$ (see Definition 3.7). We claim that I is generated by those $f^n - f$ with $\deg(f) < n$. In fact, if $f \in \mathbb{F}_p[T]$ is arbitrary, then polynomial division yields a polynomial f' of degree $< n$ with $f \equiv f' \bmod T^n - T$, which also implies $f^n - f \equiv f'^n - f' \bmod T^n - T$. Since $\mathbb{F}_p[T]/I$ is a p^k-ring by Corollary 3.5, we have $T^{p^k} - T \in I = \langle g \rangle$, which proves $g \mid T^{p^k} - T$.

Since we already wrote g as a linear combination of polynomials of the form $f^n - f$, we can now do the same for $T^{p^k} - T$. So, we have found an equation of the form

$$T^{p^k} - T = \sum_{i=1}^{s} u_i \cdot (f_i^n - f_i)$$

in $\mathbb{F}_p[T]$. Now, if A is any ring with $p = 0$, then for $x \in A$ we get

$$x^{p^k} - x = \sum_{i=1}^{s} u_i(x) \cdot (f_i(x)^n - f_i(x)).$$

For a pure equational proof that is independent from the proof above, just expand the right hand side of this equation and then simplify. The result will be $x^{p^k} - x$. If A is an n-ring with $p = 0$, the right hand side vanishes, so that $x^{p^k} = x$.

Finally, if $p^k - 1 \mid n - 1$, then every p^k-ring is an n-ring by Corollary 2.10. □

Corollary 4.2. *In order to give an equational proof that n-rings are commutative, one may restrict to p^k-rings with $p = 0$.*

Proof. This follows from Theorem 3.14 and Theorem 4.1. □